Homework 4 — changed numbers (extra practice)

Module 4 - Drug Elimination and Clearance · Homework · No date

The numbers on this sheet are changed from her sheet.

Name _____ Date _____

1. Using IBW, estimate the creatinine clearance for a 38-year-old female patient who is 163 cm tall, weighs 58 kg, and has a serum creatinine of 0.92 mg/dL.

Problem 1 What is her IBW?

Problem 1 What is the estimated CrCl? (First part: IBW = 54.7 kg.)

2. A drug with an elimination half-life of approximately 6 hours and an apparent volume of distribution of 0.3 L/kg was administered to a male volunteer (70 kg) by a

bolus intravenous injection at a dose level of 15 mg/kg.

2a What is the expected plasma concentration 10 hours after administration of the dose?

2b What is the expected AUC in this patient following this IV bolus dose?

2c Urine samples were collected and analyzed for 48 hours following the dose. If 735 mg of drug was recovered in the urine, what is the excretion rate constant of this agent?

2d Urine samples were collected and analyzed for 48 hours following the dose. 735 mg of drug was recovered in the urine. d. What is the renal clearance of this drug?

2e Urine samples were collected and analyzed for 48 hours following the dose. 735 mg of drug was recovered in the urine. e. What is the probable mechanism for renal clearance of this drug?

A. Active secretion only
B. Filtration only
C. Filtration with active secretion
D. Filtration with tubular reabsorption

2f Urine samples were collected and analyzed for 48 hours following the dose. 735 mg of drug was recovered in the urine. f. What is the hepatic clearance of this drug in this patient?

2g If this drug were administered to a patient with partial renal failure and a clearance of 1.2 L/hr (assuming no change to the volume of distribution), what elimination half-life would you expect?

2h What dose would you recommend to provide the same AUC (as determined in part b) for this patient with partial renal failure and clearance of 1.2L/hr?